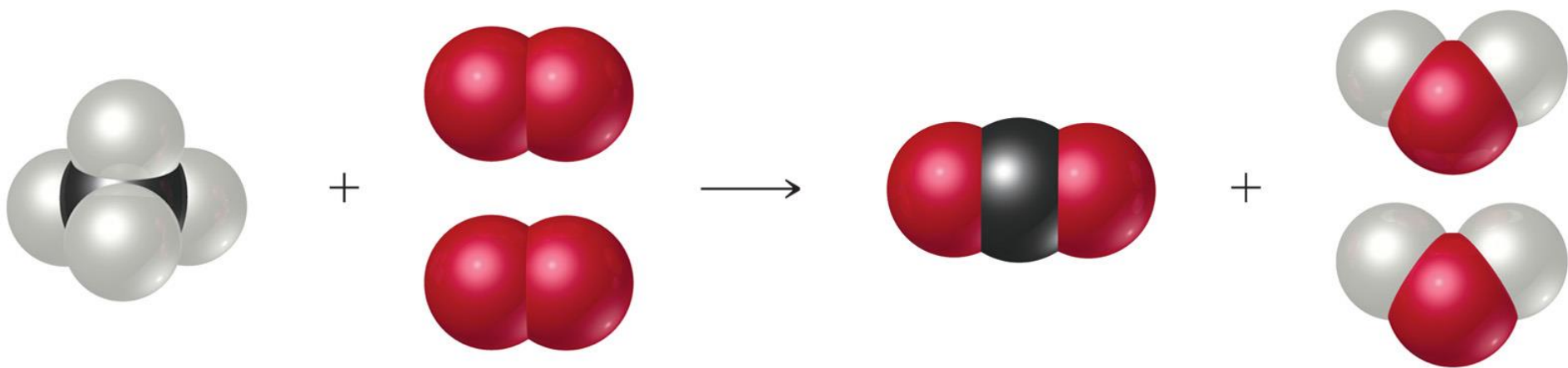
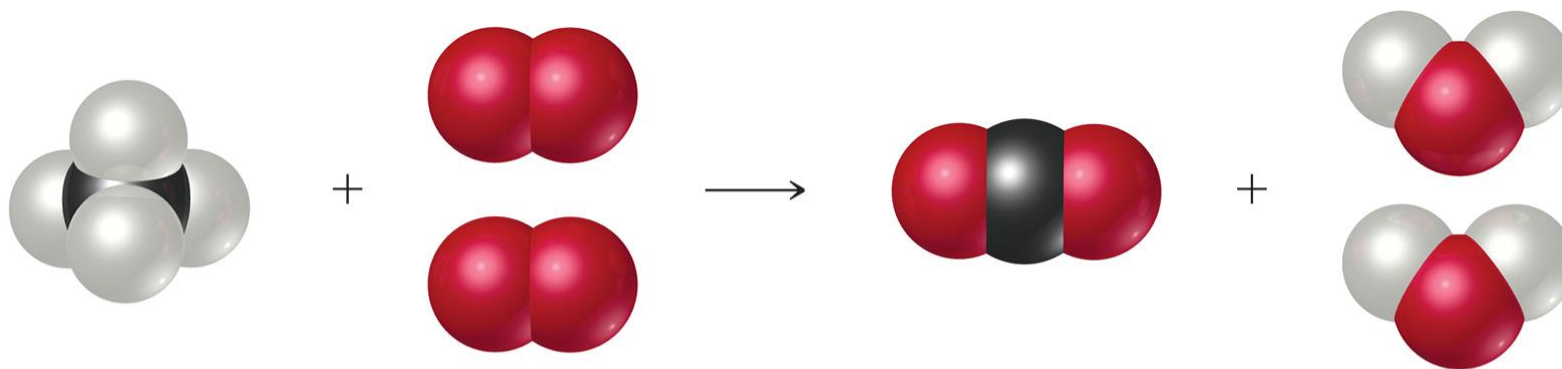


Chemical Equations

Chemical equations are concise representations of chemical reactions.



Anatomy of a Chemical Equation



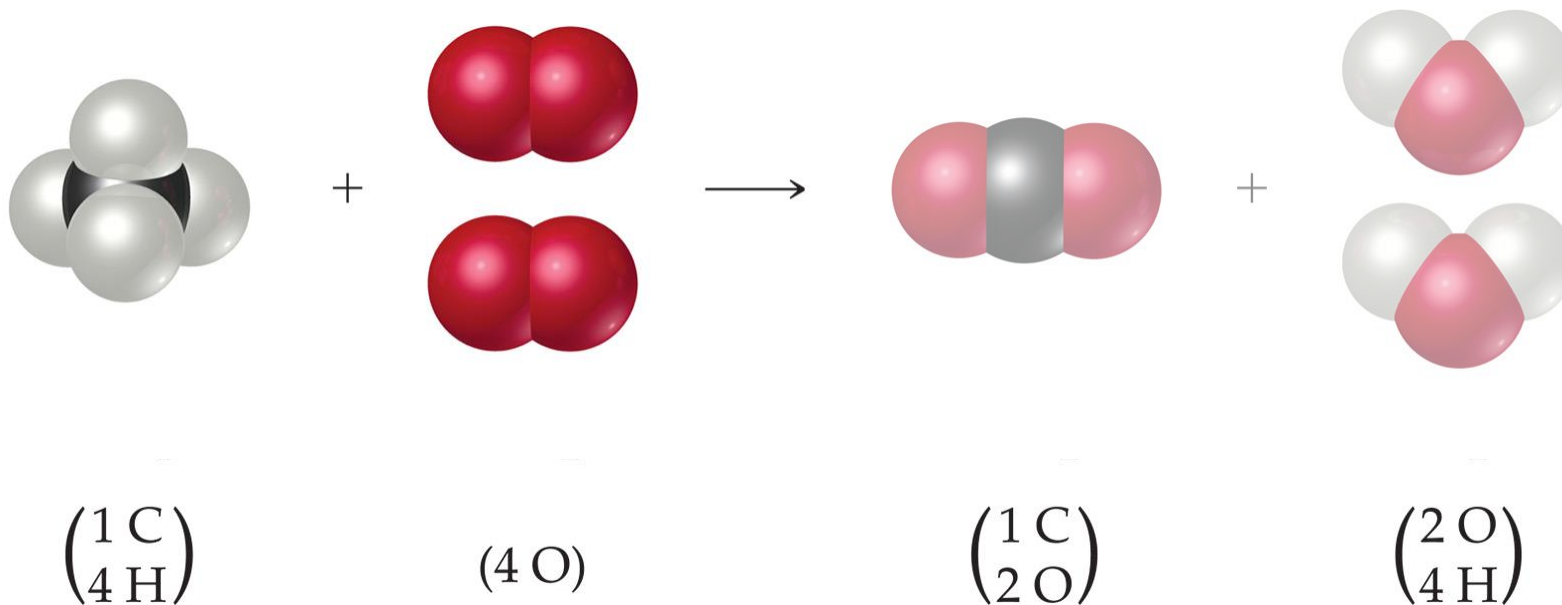
$\begin{pmatrix} 1 \text{ C} \\ 4 \text{ H} \end{pmatrix}$

(4 O)

$\begin{pmatrix} 1 \text{ C} \\ 2 \text{ O} \end{pmatrix}$

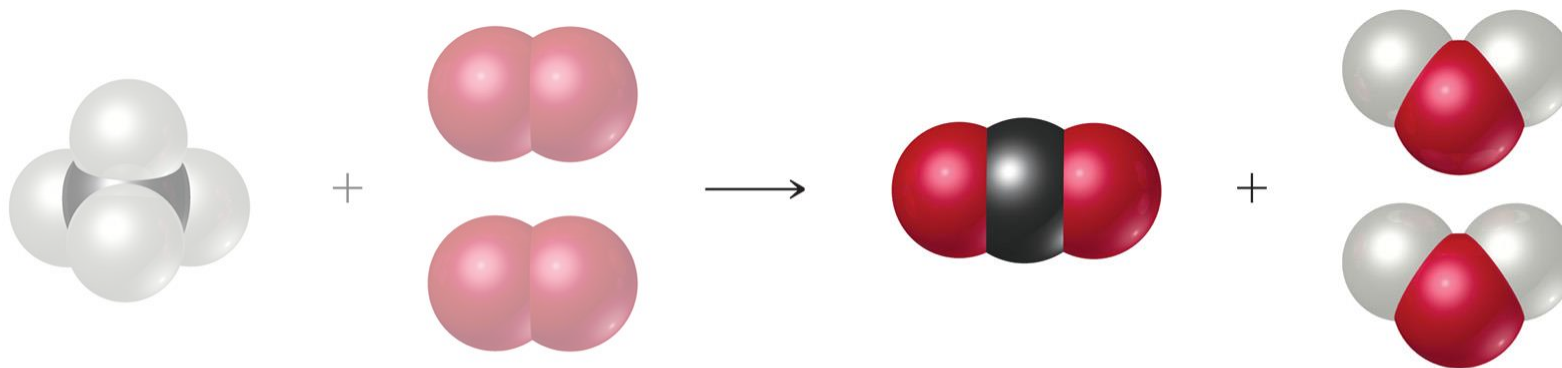
$\begin{pmatrix} 2 \text{ O} \\ 4 \text{ H} \end{pmatrix}$

Anatomy of a Chemical Equation



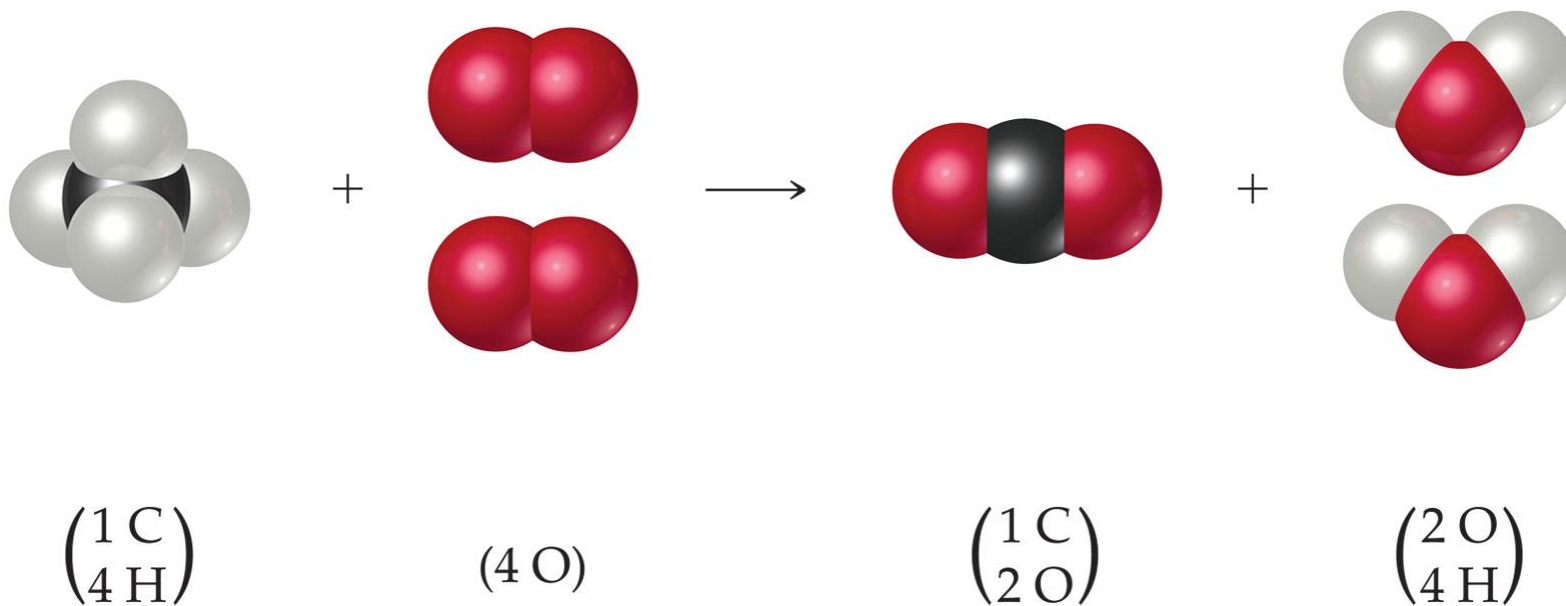
Reactants appear on the left side of the equation.

Anatomy of a Chemical Equation



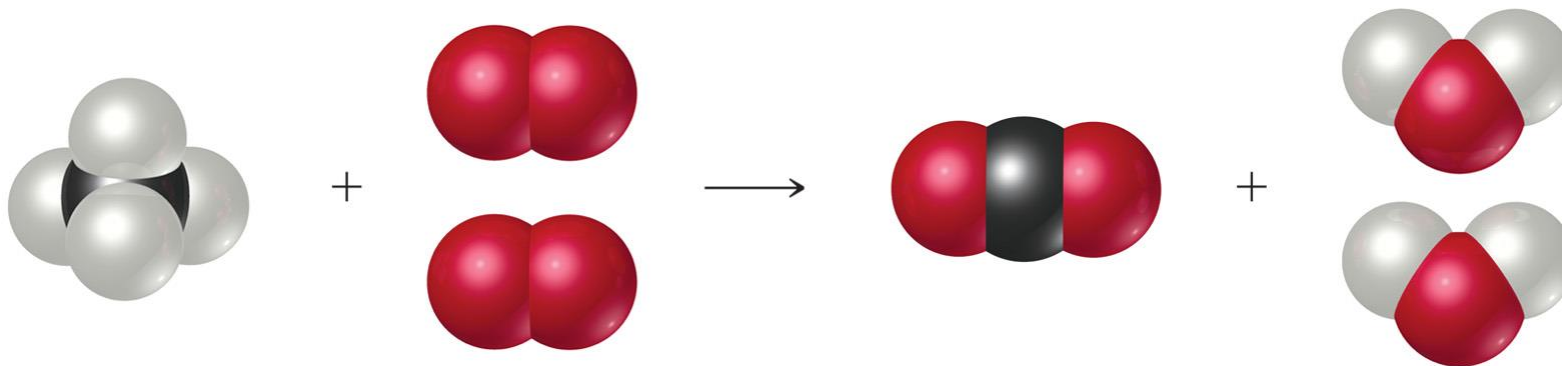
Products appear on the right side of the equation.

Anatomy of a Chemical Equation



The **states** of the reactants and products are written in parentheses to the right of each compound.

Anatomy of a Chemical Equation



Coefficients are inserted
to balance the equation.

Subscripts and Coefficients Give Different Information

Chemical symbol	Meaning	Composition
H_2O	One molecule of water:	Two H atoms and one O atom
$2 \text{H}_2\text{O}$	Two molecules of water:	Four H atoms and two O atoms

- Subscripts tell the number of atoms of each element in a molecule.

Subscripts and Coefficients Give Different Information

Chemical symbol	Meaning	Composition
H_2O	One molecule of water:	Two H atoms and one O atom
$2 \text{H}_2\text{O}$	Two molecules of water:	Four H atoms and two O atoms

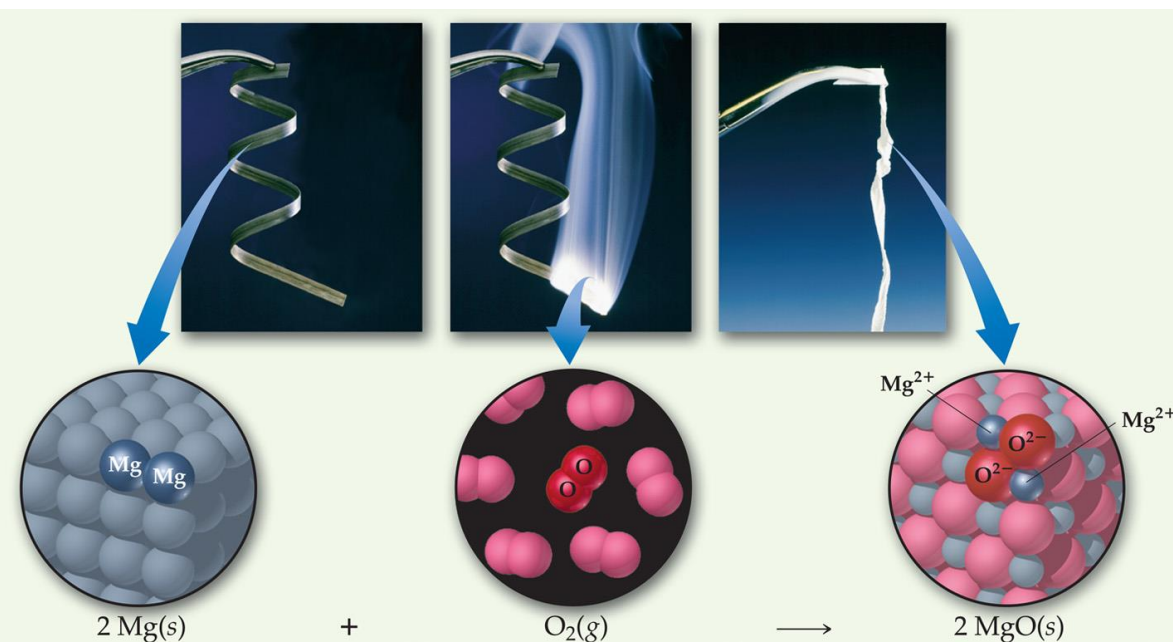
- Subscripts tell the number of atoms of each element in a molecule
- Coefficients tell the number of molecules.



Reaction Types



Combination Reactions

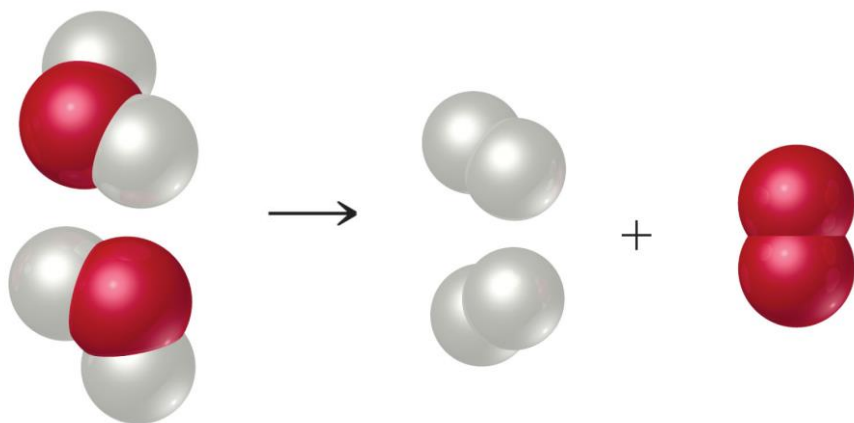


- In this type of reaction two or more substances react to form one product.

- Examples:

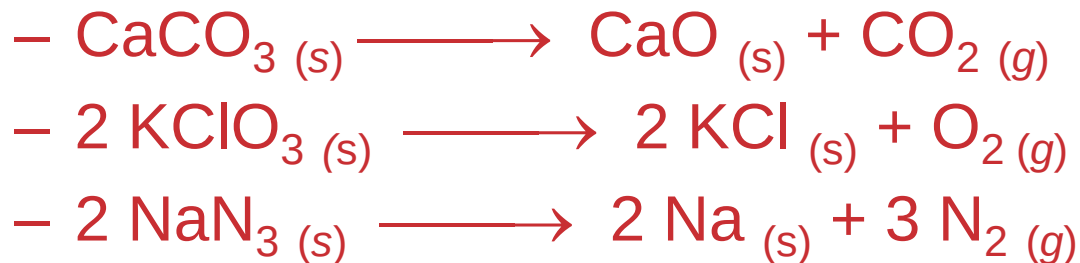


Decomposition Reactions

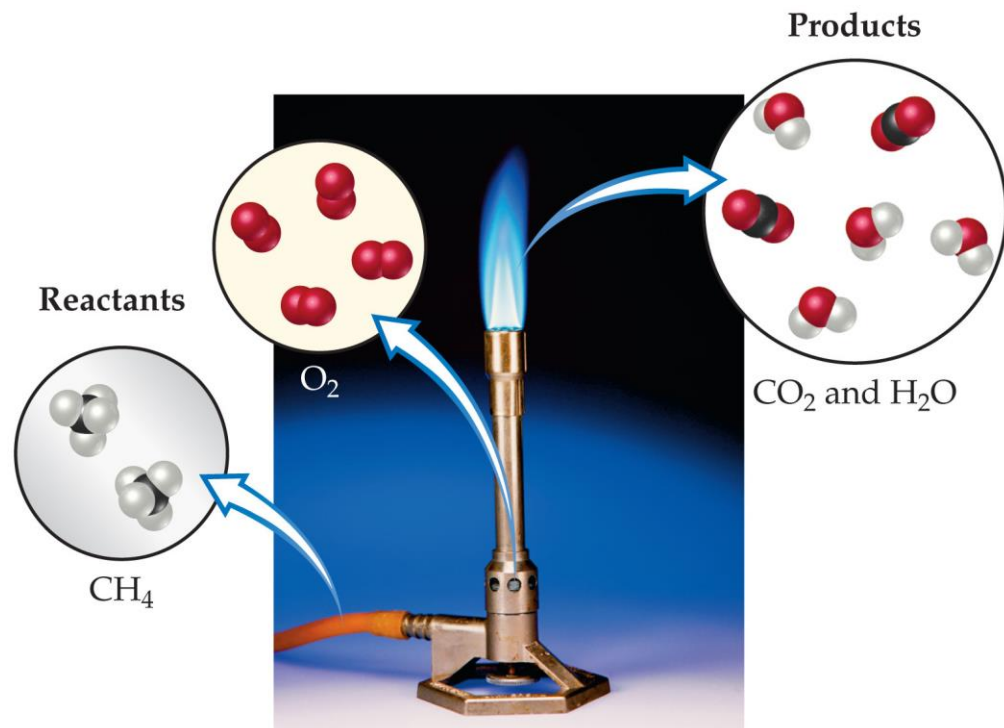


- In a decomposition one substance breaks down into two or more substances.

- Examples:



Combustion Reactions



- These are generally rapid reactions that produce a flame.
- Most often involve hydrocarbons reacting with oxygen in the air.

- Examples:



Formula Weights



Formula Weight (FW)

- A formula weight is the sum of the atomic weights for the atoms in a chemical formula.
- So, the formula weight of calcium chloride, CaCl_2 , would be

$$\begin{array}{r} \text{Ca: } 1(40.1 \text{ amu}) \\ + \text{ Cl: } \underline{2(35.5 \text{ amu})} \\ 111.1 \text{ amu} \end{array}$$

- Formula weights are generally reported for ionic compounds.



Molecular Weight (MW)

- A molecular weight is the sum of the atomic weights of the atoms in a molecule.
- For the molecule ethane, C_2H_6 , the molecular weight would be

$$\begin{array}{r} \text{C: } 2(12.0 \text{ amu}) \\ + \text{ H: } 6(1.0 \text{ amu}) \\ \hline 30.0 \text{ amu} \end{array}$$



Percent Composition

One can find the percentage of the mass of a compound that comes from each of the elements in the compound by using this equation:

$$\% \text{ element} = \frac{(\text{number of atoms})(\text{atomic weight})}{(\text{FW of the compound})} \times 100$$



Percent Composition

So the percentage of carbon in ethane is...

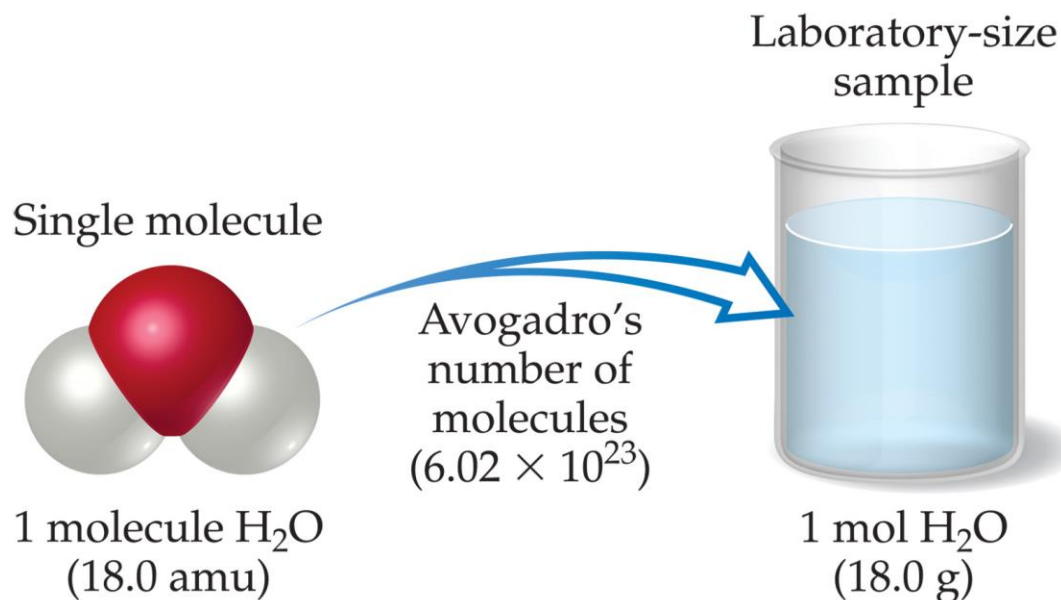
$$\begin{aligned}\%C &= \frac{(2)(12.0 \text{ amu})}{(30.0 \text{ amu})} \\ &= \frac{24.0 \text{ amu}}{30.0 \text{ amu}} \times 100 \\ &= 80.0\%\end{aligned}$$



Moles



Avogadro's Number



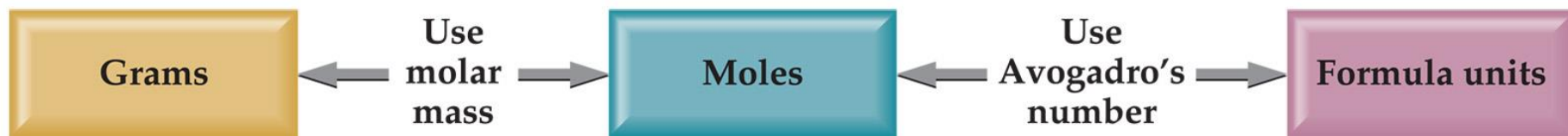
- 6.02×10^{23}
- 1 mole of ^{12}C has a mass of 12 g.

Molar Mass

- By definition, a molar mass is the mass of 1 mol of a substance (i.e., g/mol).
 - The molar mass of an element is the mass number for the element that we find on the periodic table.
 - The formula weight (in amu's) will be the same number as the molar mass (in g/mol).



Using Moles



Moles provide a bridge from the molecular scale to the real-world scale.



Mole Relationships

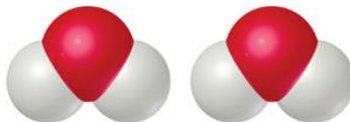
Name of Substance	Formula	Formula Weight (amu)	Molar Mass (g/mol)	Number and Kind of Particles in One Mole
Atomic nitrogen	N	14.0	14.0	6.02×10^{23} N atoms
Molecular nitrogen	N ₂	28.0	28.0	$\left\{ \begin{array}{l} 6.02 \times 10^{23} \text{ N}_2 \text{ molecules} \\ 2(6.02 \times 10^{23}) \text{ N atoms} \end{array} \right.$
Silver	Ag	107.9	107.9	6.02×10^{23} Ag atoms
Silver ions	Ag ⁺	107.9 ^a	107.9	6.02×10^{23} Ag ⁺ ions
Barium chloride	BaCl ₂	208.2	208.2	$\left\{ \begin{array}{l} 6.02 \times 10^{23} \text{ BaCl}_2 \text{ units} \\ 6.02 \times 10^{23} \text{ Ba}^{2+} \text{ ions} \\ 2(6.02 \times 10^{23}) \text{ Cl}^- \text{ ions} \end{array} \right.$

^a Recall that the electron has negligible mass; thus, ions and atoms have essentially the same mass.

- One mole of atoms, ions, or molecules contains Avogadro's number of those particles.
- One mole of molecules or formula units contains Avogadro's number times the number of atoms or ions of each element in the compound.



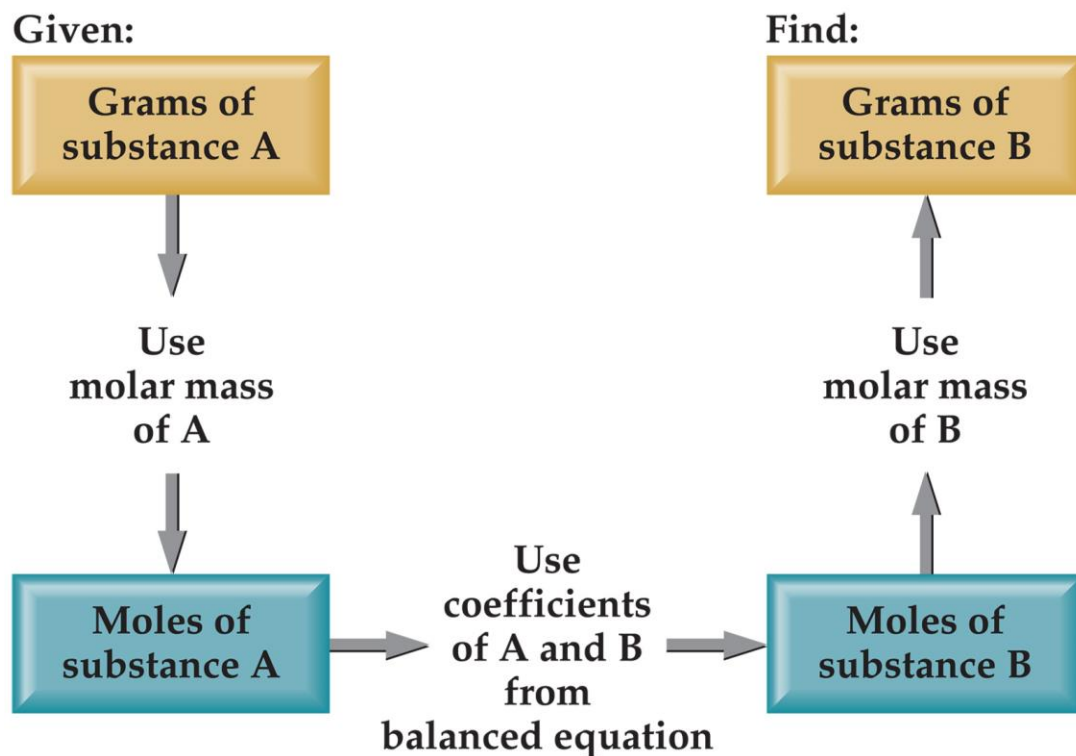
Stoichiometric Calculations

Equation:	$2 \text{H}_2(\text{g})$	+	$\text{O}_2(\text{g})$	$\longrightarrow$	$2 \text{H}_2\text{O}(\text{l})$
Molecules:	2 molecules H_2	+	1 molecule O_2	$\longrightarrow$	2 molecules H_2O
					
Mass (amu):	4.0 amu H_2	+	32.0 amu O_2	$\longrightarrow$	36.0 amu H_2O
Amount (mol):	2 mol H_2	+	1 mol O_2	$\longrightarrow$	2 mol H_2O
Mass (g):	4.0 g H_2	+	32.0 g O_2	$\longrightarrow$	36.0 g H_2O

The coefficients in the balanced equation give the ratio of *moles* of reactants and products.

Stoichiometric Calculations

Starting with the mass of Substance A you can use the ratio of the coefficients of A and B to calculate the mass of Substance B formed (if it's a product) or used (if it's a reactant).



Stoichiometric Calculations



1.00 g $\text{C}_6\text{H}_{12}\text{O}_6$
--

Starting with 1.00 g of $\text{C}_6\text{H}_{12}\text{O}_6$...
we calculate the moles of $\text{C}_6\text{H}_{12}\text{O}_6$...
use the coefficients to find the moles of H_2O ...
and then turn the moles of water to grams.



Limiting Reactants



How Many Cookies Can I Make?



- You can make cookies until you run out of one of the ingredients.
- Once this family runs out of sugar, they will stop making cookies (at least any cookies you would want to eat).



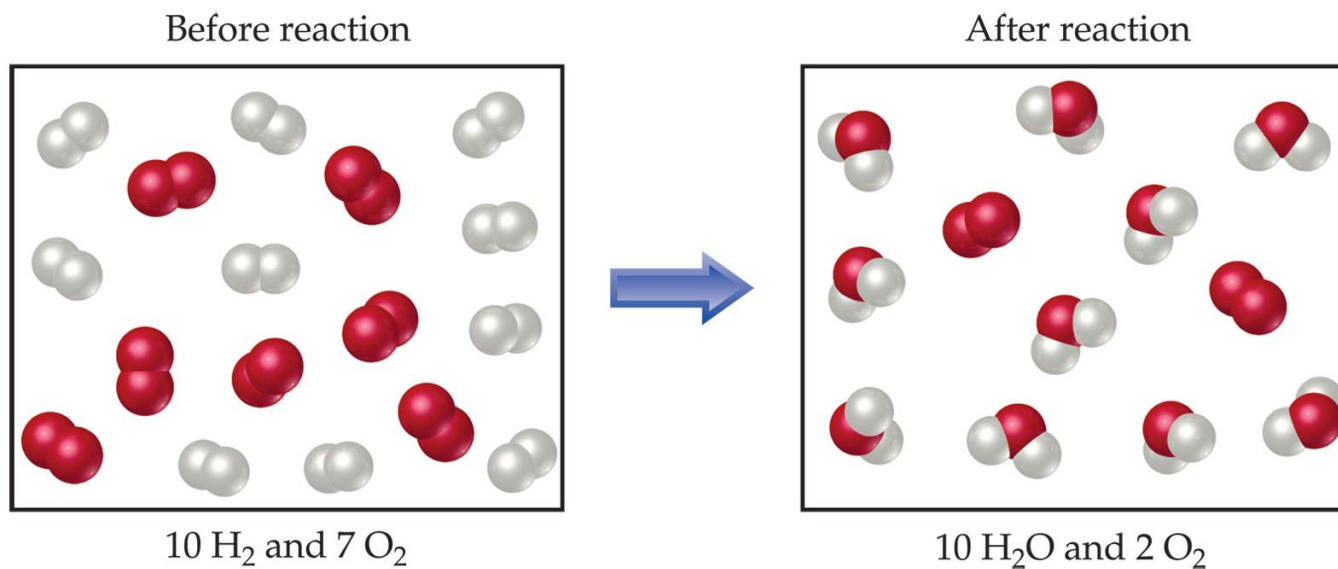
How Many Cookies Can I Make?



- In this example the sugar would be the limiting reactant, because it will limit the amount of cookies you can make.

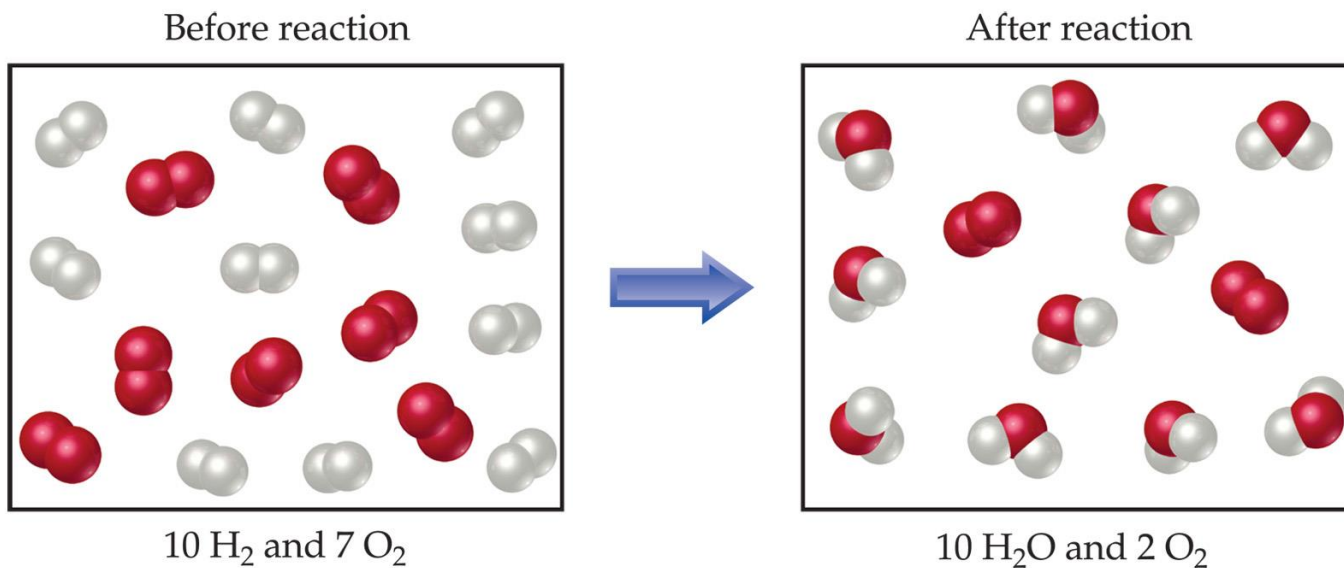
Limiting Reactants

- The limiting reactant is the reactant present in the smallest stoichiometric amount.
 - In other words, it's the reactant you'll run out of first (in this case, the H_2).



Limiting Reactants

In the example below, the O_2 would be the excess reagent.



Theoretical Yield

- The theoretical yield is the maximum amount of product that can be made.
 - In other words it's the amount of product possible as calculated through the stoichiometry problem.
- This is different from the actual yield, which is the amount one actually produces and measures.



Percent Yield

One finds the percent yield by comparing the amount actually obtained (actual yield) to the amount it was possible to make (theoretical yield).

$$\text{Percent Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$



Concentrations of Solutions

- A solution is a homogenous mixture of two or more substances (solute and solvent).
- The solvent is the major component.
- The solute is the minor component.
- The behavior of solutions often depends on the nature of the solutes and their concentration.



Ways to express concentration

- Qualitative:- Express the concentrations of two solutions relatively.
- Quantitative:- Express the exact concentration of a solution.
 - Molarity.
 - Molality.
 - Mass percent.
 - Mole fraction.



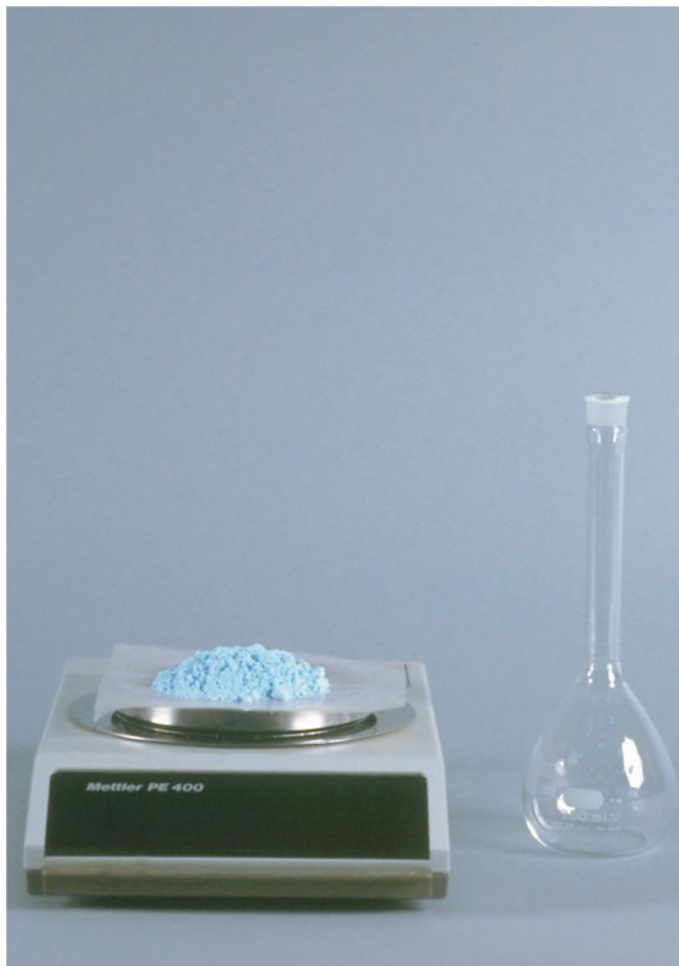
Molarity

- Two solutions can contain the same compounds but be quite different because the proportions of those compounds are different.
- Molarity is one way to measure the concentration of a solution.

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{volume of solution in liters}}$$



Mixing a Solution



- To create a solution of a known molarity, one weighs out a known mass (and, therefore, number of moles) of the solute.
- The solute is added to a volumetric flask, and solvent is added to the line on the neck of the flask.

Dilution

- One can also dilute a more concentrated solution by
 - Using a pipet to deliver a volume of the solution to a new volumetric flask, and
 - Adding solvent to the line on the neck of the new flask.



Dilution

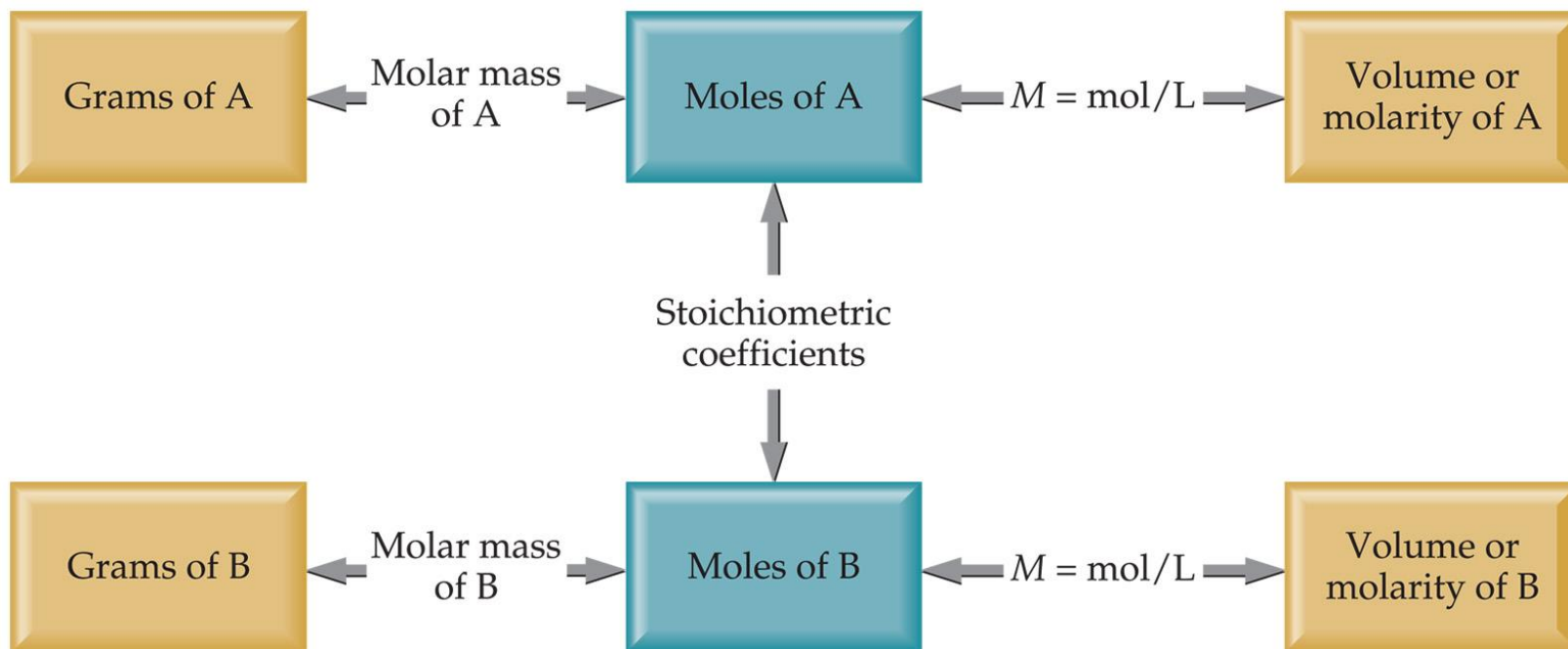
The molarity of the new solution can be determined from the equation

$$M_c \times V_c = M_d \times V_d,$$

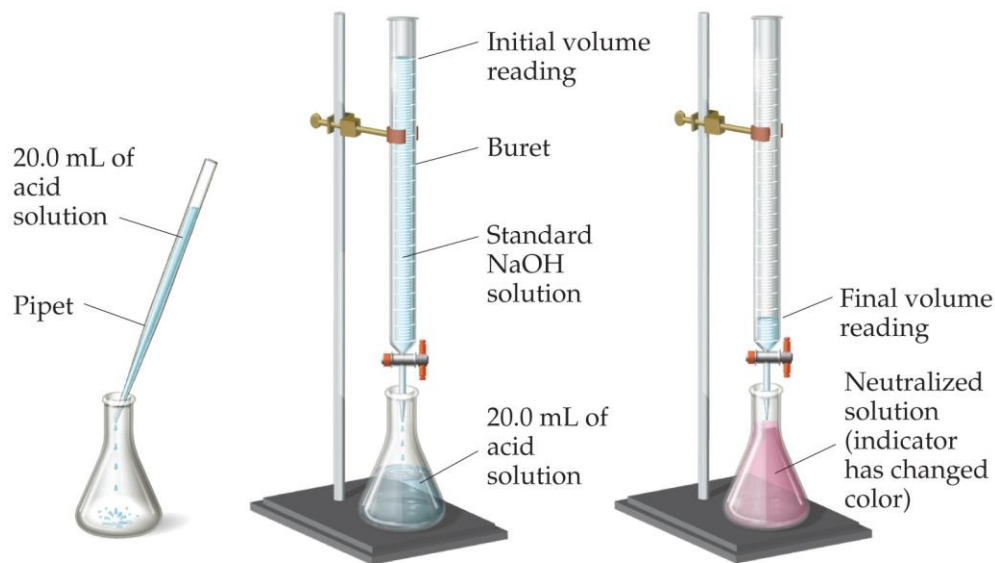
where M_c and M_d are the molarity of the concentrated and dilute solutions, respectively, and V_c and V_d are the volumes of the two solutions.



Using Molarities in Stoichiometric Calculations



Titration



Titration is an analytical technique in which one can calculate the concentration of a solute in a solution.

Titration Law

$$M_a V_a = M_b V_b$$

Where : a = acid and b= base.

An indicator is used to indicate the equivalence point and the end point

